Java Keywords & Identifiers

1.Introduction

Java is one of the most popular programming languages in the world. It is widely used for building web applications, enterprise software, Android apps, and large-scale systems.

Before writing complex programs in Java, it is important to understand the basic building blocks of the language.

Two of the most important concepts in Java are:

- Keywords
- Identifiers

These two elements form the foundation of Java syntax.

Every Java program you write will contain identifiers, and every program will rely on keywords defined by the Java language.

Understanding them clearly will help you:

- Write clean and correct Java programs
- Avoid syntax errors
- Understand Java code easily
- Follow good coding practices

In this article, we will deeply understand Java Keywords and Identifiers with examples.

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2. What are Keywords in Java?

Definition

Keywords in Java are reserved words that have a predefined meaning in the Java programming language.

These words are part of the Java syntax, and they cannot be used as names for variables, classes, methods, or identifiers.

In simple words:

👉 Keywords are special words that Java already uses for its own functionality.

Example

int number = 10;

Here:

- int → keyword
- number → identifier

Why Java Keywords Exist

Keywords help Java define:

- data types
- control statements
- loops
- class structures
- object-oriented features

Without keywords, Java would not understand how a program should behave.

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3. Characteristics of Java Keywords

Java keywords have the following properties:

1. Reserved Words

Keywords are reserved for Java language syntax.

Example:

int class = 10; // ❌ invalid

Because class is a keyword.

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2. Cannot be used as identifiers

You cannot use keywords as:

- variable names
- method names
- class names
- object names

Example:

int if = 20; // ❌ not allowed

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3. Lowercase in Java

Most Java keywords are written in **lowercase**.

Example:

class
public
static
void

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4. Fixed Meaning

Each keyword has a **specific meaning defined by Java**.

You cannot change its behavior.

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4. Total Number of Keywords in Java

Java currently has 50 reserved keywords.

Examples include:

abstract
assert
boolean
break
byte
case
catch
char
class
const
continue
default
do
double
else
enum
extends
final
finally
float
for
goto
if
implements

import
instanceof
int
interface
long
native
new
package
private
protected
public
return
short
static
strictfp
super
switch
synchronized
this
throw
throws
transient
try
void
volatile
while

Note:

const and goto are reserved but not used.

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5. Categories of Java Keywords

Java keywords can be grouped based on their purpose.

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6. Data Type Keywords

These keywords define primitive data types.

Examples:

byte
short
int
long
float
double
char
boolean

Example program:

```
public class DataTypeExample {

   public static void main(String[] args) {

      int age = 25;
      double salary = 50000.50;
      char grade = 'A';
      boolean isPassed = true;

      System.out.println(age);
      System.out.println(salary);
      System.out.println(grade);
      System.out.println(isPassed);

   }
}
```

---

7. Access Modifier Keywords

These keywords control visibility of classes, methods, and variables.

Examples:

public
private
protected

Example:

```
public class Student {

   private int id;
   protected String name;

}
```

---

8. Control Flow Keywords

These keywords control program execution flow.

Examples:

if
else
switch
case
default
break
continue

Example:

```
int number = 10;

if(number > 0){
   System.out.println("Positive number");
```

```
}
else{
   System.out.println("Negative number");
}
```

---

9. Loop Keywords

Loops allow repeating a block of code.

Examples:

for
while
do

Example:

```
for(int i = 1; i <= 5; i++){

   System.out.println(i);

}
```

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10. Class Related Keywords

These keywords are used in object-oriented programming.

Examples:

class
extends
implements
interface
super
this
new

Example:

```
class Animal{

   void sound(){
      System.out.println("Animal makes sound");
   }

}
```

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11. Exception Handling Keywords

Java provides special keywords for handling errors.

Examples:

try
catch
finally
throw
throws

Example:

```
try{

   int result = 10 / 0;

}
catch(Exception e){

   System.out.println("Exception occurred");

}
```

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12. Other Important Keywords

Some other important keywords include:

static
final
abstract
synchronized
volatile
native
transient

Example:

final int number = 10;

Here the value cannot be changed.

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13. What are Identifiers in Java?

Definition

An identifier is the name given to a variable, method, class, or object in Java.

Identifiers are used to identify programming elements.

Example

int age = 25;

Here:

- int → keyword
- age → identifier

---

14. Types of Identifiers in Java

Identifiers can represent:

1. Class Names

class Student

Student is an identifier.

---

2. Variable Names

int marks;

marks is an identifier.

---

3. Method Names

void display()

display is an identifier.

---

4. Package Names

package com.example;

example is an identifier.

---

15. Rules for Identifiers in Java

Java has strict rules for naming identifiers.

Rule 1: Must start with letter, underscore, or $

Valid examples:

name
_age
$salary

Invalid:

1name

Rule 2: Cannot start with number

int 1number; ❌

Rule 3: Cannot use Java keywords

int class = 10; ❌

Rule 4: No special symbols

Invalid:

int total-amount ❌

Rule 5: Case Sensitive

age
Age
AGE

All are different identifiers.

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16. Examples of Valid Identifiers

name
studentId
totalMarks
salaryAmount

---

17. Examples of Invalid Identifiers

1name
class
total-marks
int

---

18. Naming Conventions in Java

Good programmers follow naming conventions.

Class Naming

Use PascalCase

StudentDetails
EmployeeData
BankAccount

Variable Naming

Use camelCase

studentName
totalMarks
accountBalance

Constant Naming

Use UPPERCASE

MAX_VALUE
MIN_VALUE

---

19. Keywords vs Identifiers

Purpoes	Keywords	Identifiers
Meaning	Reserved words	Names given by programmer
Usage	Define language syntax	Identify variables, classes
Example	int, class, public	age, student, total

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20. Example Program Using Keywords & Identifiers

```
public class Student {

  int id;
  String name;

  public void display(){

    System.out.println(id);
    System.out.println(name);

  }
```

```
    public static void main(String[] args){

        Student s1 = new Student();

        s1.id = 101;
        s1.name = "Omkar";

        s1.display();
    }
}
```

Keywords used:

public
class
int
void
static
new

Identifiers used:

Student
id
name
display
s1

---

21. Common Mistakes Beginners Make

1️⃣ Using keywords as variable names

```
int class = 10;
```

2️⃣ Starting identifiers with numbers

```
int 1value;
```

3️⃣ Using spaces in identifiers

int total marks;

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22. Best Practices

✓ Use meaningful names
✓ Follow camelCase naming
✓ Avoid very long names
✓ Keep identifiers readable

Example:

Good:

studentAge

Bad:

sa

---

23. Conclusion

Keywords and identifiers are two fundamental elements of Java programming.

Keywords define the structure and rules of the language, while identifiers allow programmers to name and manage program elements.

Understanding these concepts clearly helps developers:

- write clean code
- avoid syntax errors
- follow good programming practices

Mastering these basics is the first step toward becoming a strong Java developer.